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Advanced Computer Architecture

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Name
Last Name
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You have 90' to complete the exam

Problem 1 (6pts)	
Problem 2 (6pts)	
Problem 3 (6pts)	
Question 1 (5pts)	
Question 2 (5pts)	
Question 3 (5pts)	
Total (32pts)	

No PCs (or other electronic devices), notes, books, are allowed during the exam.

To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem and 1 Question has to be correct

Problem 1

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	LD F5, 16 (R1)	1	2	6	7
I2	ADDD F12, F5, F2	2	8	14	15
I3	MULTD F2, F4, F3	3	4	15	16
I4	DIVD F1, F12, F5	4	16	27	28
I5	SD F1, 4 (R1)	5	29	33	34
I6	SUBD F2, F12, F4	17	18	24	25

A. List all the possible conflicts in the code.

B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

C. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

Answer 3.A

RAW F5 I1-I2
RAW F5 I1-I4
RAW F12 I2-I4
RAW F12 I2-I6
RAW F1 I4-I5

WAW F2 I3-I6

WAR F2 I2-I3
WAR F2 I2-I6

Answer 3.B

A possible configuration consists of:

- 2 MUL/DIV units and 11 CC of latency
- 1 ADDD/SUBD unit and 6 CC of latency
- 1 Memory Unit and 4 CC of latency
- 1 write port is enough

Other solutions are possible.

Answer 3.C

Considering the previous answers, no answer is needed here.

Problem 2

Consider two CPUs: CPU1 e CPU2. CPU1 has clock cycle of 2 ns while CPU2 has an operating frequency of 700MHz. Given the following frequencies of occurrence of the instructions for the two CPUs

Operation type	Frequency	CPU1	CPU2
A	0.3	2	2
B	0.1	3	3
C	0.2	4	3
D	0.3	2	2
E	0.1	4	3

- a) Compute the average CPI for CPU1 and CPU2
b) Which is the slowest CPU?
- c) Assume that CPU1 is modified in such a way that:
- All instructions require 8 clock cycles;
 - Memory stalls are not considered;
 - The miss penalty is 7 clock cycles

Assuming a miss rate of 23% and assuming 3 memory references on average for each instruction determine the impact on performance of the cache with respect to an ideal cache?

Answer

a)

$$\text{CPI-CPU1} = 0.3 \cdot 2 + 0.1 \cdot 3 + 0.2 \cdot 4 + 0.3 \cdot 2 + 0.1 \cdot 4 = 0.6 + 0.3 + 0.8 + 0.6 + 0.4 = \mathbf{2.7}$$

$$\text{CPI-CPU2} = 0.3 \cdot 2 + 0.1 \cdot 3 + 0.2 \cdot 3 + 0.3 \cdot 2 + 0.1 \cdot 3 = 0.6 + 0.3 + 0.6 + 0.6 + 0.3 = \mathbf{2.4}$$

b) Speedup = Exe-CPU1/Exe-CPU2 =

$$\begin{aligned} &= ((IC \cdot CPI1)/Freq1) \cdot (Freq2/(IC \cdot CPI2)) = \\ &= (IC \cdot CPI1 \cdot Freq2)/(IC \cdot CPI2 \cdot Freq1) = \\ &= (CPI1 \cdot Freq2)/(CPI2 \cdot Freq1) = \\ &= (2.7 \cdot 700\text{MHz})/(2.4 \cdot 500\text{MHz}) = 1890/1200 = \mathbf{1.575} \end{aligned}$$

c)

CPI_exe = 8; MISS Penalty = 7; MISS Rate = 0.23; Memory references = 3

CPI_ideal_cache (MISS Rate = 0) = **8**

$$\begin{aligned} \text{CPI_cache} &= \text{CPI_exe} + (\text{References} \cdot \text{MISSPenalty} \cdot \text{MISSRate}) = \\ &= 8 + (3 \cdot 7 \cdot 0.23) = 8 + 4.83 = \mathbf{12.83} \end{aligned}$$

Problem 3

Details about the **4-issue VLIW** machine with 4 fully pipelined multicycles functional units:

- Integer ALU with 1 cycle latency
- 1 Memory Unit with 3 cycle latency
- 2 Floating Point Unit with 3 cycle latency (it can complete one add or one multiply per clock cycle)
- Branch completed with 1 cycle delay slot (branch solved in ID stage)
- In the Register File, it is possible to read and write at the same address at the same clock cycle

Considering the following portion of assembly code:

```
Loop:      LD F0, 0(R1)
           LD F2, 4(R1)
           LD F3, 8(R1)
           LD F4, 12(R1)
           FADD F5, F0, F1
           FADD F6, F2, F1
           FADD F7, F3, F1
           FADD F8, F4, F1
           FMULT F9, F0, F1
           FMULT F10, F2, F1
           FMULT F11, F3, F1
           FMULT F12, F4, F1
           FADD F13, F9, F5
           FADD F14, F10, F6
           FADD F15, F11, F7
           FADD F16, F12, F8
           SD F13, 0(R1)
           SD F14, 4(R1)
           SD F15, 8(R1)
           SD F16, 12(R1)
           ADD R1, R1, 16
           BNE R1, R2, Loop
```

Schedule one iteration of the loop on the 4-issue VLIW machine (we don't write the NOPs) with and IN-ORDER ISSUE and pipelined FUs.

	Int ALU	Mem Unit	FPU+	FPUx
C0		LD F0, 0(R1)		
C1		LD F2, 4(R1)		
C2		LD F3, 8(R1)		
C3		LD F4, 12(R1)	FADD F2, F0, F1	FMULT F9, F0, F1
C4			FADD F6, F2, F1	FMULT F10, F2, F1
C5			FADD F7, F3, F1	FMULT F11, F3, F1
C6			FADD F8, F4, F1	FMULT F12, F4, F1
C7			FADD F13, F9, F5	
C8			FADD F14, F10, F6	
C9			FADD F15, F11, F7	
C10		SD F13, 0(R1)	FADD F16, F12, F8	
C11		SD F14, 4(R1)		
C12		SD F15, 7(R1)		
C13	ADD R1,R1,16	SD F16, 12(R1)		
C14	BNE R1 R2 Loop			
C15	Br delay slot			
C16				
C17				
C18				
C19				
C20				
C21				
C22				
C23				
C24				
C25				
C26				
C27				

Question 1

The following 5 questions are True/False (you do not have to motivate/support your answers).

Each of them will count 1 point

Question 1.1

Using the "Early Evaluation of PC" applied to branch instructions, we need 3 stalls to execute the following assembly code

```
lw $1, 0($5)
beq $1, $0, label
```

Circle the **right** answer: True False

Answer 1.1

FALSE! If a comparison register is a destination of immediately preceding load instruction 2 stall cycles are needed in order to allow MEM/ID forwarding of the result of the load.

Question 1.2

TS is particularly useful in a code with many loops without control-flow

Circle the **right** answer: True False

Answer 1.2

FALSE! TS is used in control-flow intensive irregular code where branches limit basic block size, and therefore it is difficult to exploit ILP in single blocks.

Question 1.3

A physically distributed memory can be used to implement a shared-memory architecture

Circle the **right** answer: True False

Answer 1.3

TRUE! Both the centralized memory and the distributed memory models can be used to implement a shared-memory architecture

Question 1.4

A VLIW Architecture has to have multiple Program Counters to load the necessary Multiple Data

Circle the **right** answer: True False

Answer 1.4

FALSE! A VLIW architecture is a SINGLE instruction architecture (the bundle is a SINGLE Very Long Instruction), therefore we need just one Program Counter

Question 1.5

Increasing the number of stages in a pipeline, it is always improving the performance

Circle the **right** answer: True False

Answer 1.5

FALSE! Increasing the number of stages in a pipeline may lead to a higher number of overlapping situations but also to a much more complex control unit with may not improve, in the end, the performance of the architecture.

Question 2

Explain the difference among fine-grained multithreading, coarse-grained multithreading and simultaneous multithreading.

Question 3

Explain the main differences between loop unrolling and software pipelining?

Answer 3

Software pipelining can be thought of as "symbolic" loop unrolling: the main difference is that loop unrolling needs to add some startup code before the loop as well as finish up code at the end.

On the other hand, loop unrolling needs to replicate the entire code of the loop. For what concern the register pressure, the amount of registers used are not related to the type of scheduling but only to the architecture used.

Due to the fact that loop unrolling at each iteration needs to repeat the computation, it needs to pay the cost of overhead and wind-down more times than software pipelining.

For what concerns the power consumption, because of the fact that they are both static scheduling techniques, the power consumption is lower wrt dynamic scheduling techniques. Hence, power consumption does not depend on the type of scheduling technique used.